

Assembling a dinosaur

By HAROLD SCARLETT

WHEN THE Houston Museum of Natural Science opens its new wing today, the museum's new dinosaur-in-residence could bear the proud legend: "Built in Houston by Amateurs."

The dinosaur's fossil skeleton, 150 million years old, was assembled and restored by an improbable trio:

• A designer and decorator, Arthur Pugh, who splashed ashore with the Marines on Iwo Jima in World War II.

• His winsome wife Karen, a former surgical nurse whose knowledge of anatomy came in handy on the dinosaur.

• A pert blonde, Mrs Linda Warfield, an aspiring sculptress who took some courses in the subject at Lamar State College in Beaumont.

Through the long Houston summer, in a drab and cluttered metal steam-bath of a building which is Pugh's workshop on the Southwest Freeway, the trio worked in damp devotion putting together this prehistoric monster.

Out in front for several weeks, there sat an old London double-decker bus. Pugh was decorating it as an advertising gimmick for a pizza parlor.

It was a crazy, surrealistic scene. A happening.

YOU PICTURE dinosaurs being assembled in a musty museum workshop, with dust motes impaled on fallow shafts of sunlight, doddering old scientists creaking up ladders and talking in whispers.

But this!

At one point in the proceedings, Pugh was flying around the country—from Rocky Mount, NC, to Washington to Houston—sitting in his airliner seat, cradling this dinosaur skull on his lap like a baby. It was too fragile to trust to the airline baggage-busters. The striking picture was heightened by Pugh's astrakhan cap and leather coat.

"I didn't lack for conversation anywhere I went on that trip," Pugh recalls.

Some paleontologists may have snickered when the Houston museum director, Dr Thomas E. Pulley, turned the dinosaur job over to Pugh. But Pugh, at 41 a wiry and kinetic man who looks busy even when standing still, had some impressive designing scalps on his belt — four pavilions at HemisFair, another one in the works for the 1970 World's Fair in Tokyo, decoration for restaurants and laundry chains, some small exhibits for the Houston museum.

Pugh also studied paleontology for a time at the University of Hawaii. He is a member of the Society of Vertebrate Paleontology. More important, perhaps, Pugh from boy-

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November 16, 1969

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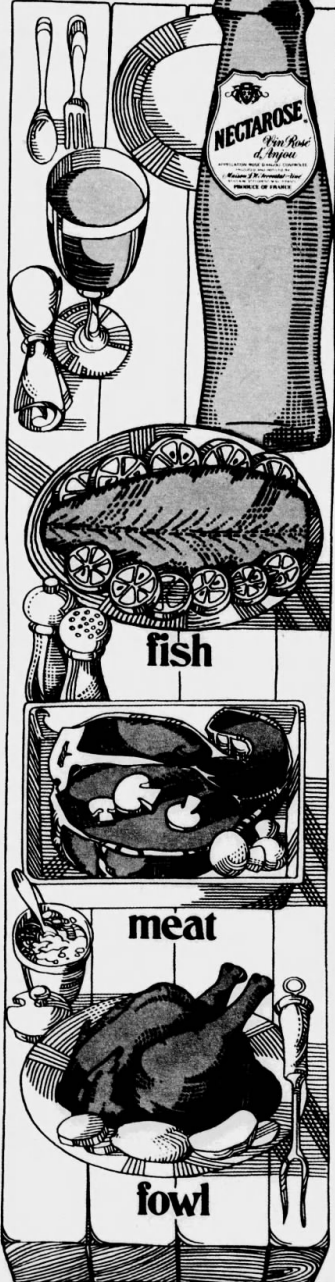
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Arthur Pugh and Linda Warfield with some pieces of the giant "jigsaw puzzle."
When finished, it will be about 78 feet long

DINOSAUR

continued

hood was caught up in the lore of dinosaurs.

He was delighted and grateful when Pulley gave him the chance to "make a lifelong dream come true."

And, after all, you won't find "Assemblers, Dinosaur" in the Yellow Pages. Very few living people in the world have ever built one.

"I was a pure amateur when we started," Pugh says. "But now, without bragging, I think I'm just as much an expert as anyone. This dinosaur is going to be as scientifically exact as we can make it."

PUGH ACQUIRED his expertise by poring over reference books, talking and corresponding with museum curators and paleontologists, assembling plates and photos and obscure publications.

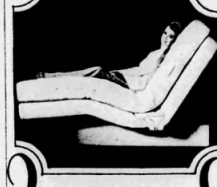
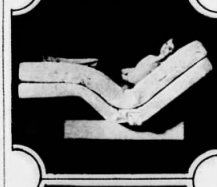
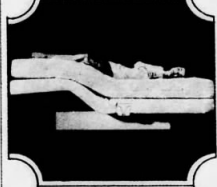
His actual working "blueprint" for assembling the Houston dinosaur was a fold-out plate from an old publication of the Carnegie Museum in Pittsburgh, a drawing of that museum's dinosaur of the same species.

The results of Pugh's project — barring some last-minute hitch—will be visible in the Alfred C. Glassell Jr Hall of the new \$2 million museum building at its formal opening today.

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For a time, the skeleton will be laid out in order on the floor of the two-story hall, just as it was originally assembled on a long trestle of tables in Pugh's workshop. It took nine months to do that. Pugh estimates it will take another six months to design and build the metal framework on which the skeleton will be mounted in the museum.

The Houston dinosaur is a diplodocus (pronounced dih-PLOD-okus), a slimmer and more graceful cousin of the brontosaurus, which is well-known as an oil company symbol.

It was a vegetarian with pencil-like front teeth to comb in the foliage. It had no chewing or fighting teeth. Grazing the ferny Jurassic swamps, it grew up to 87 feet in length and weighed perhaps 30 or 40 tons.

The Houston specimen is smaller than that, leading Mrs Pugh to conjecture that it might have been a female. No way has yet been found to tell from a skeleton whether a dinosaur was male or female.

"Ours will be about 78 feet long," Pugh says. "When mounted it will measure about 15 feet high at the shoulders, and its neck and head will arch up about 32 feet above the floor."

SO FAR AS Dr Pulley knows, there are only three other diplodocus skeletons on display in this country—at the Smithsonian in Washington, the Carnegie Museum in Pittsburgh, and the Denver Museum.

This will be the first diplodocus to be assembled anywhere from fossil bones in quite a few years. Pugh thinks it may be the first in half a century.

The Houston skeleton was discovered and excavated in 1902 on the Red Fork of the Powder River in Wyoming. It was found by Field Party 662, headed by Dr W. H. Utterbach, chief paleontologist of the Carnegie Museum.

It became Dinosaur 662, and that number—painted on each bone by the field party—will still be visible on parts of the skeleton in the Houston museum.

The field trip was one of scores financed by Andrew Carnegie, the canny Scot steel millionaire, after he became fascinated with dinosaurs.

A diplodocus for the Pittsburgh museum was assembled by combining the bones of two dinosaurs, Numbers 84 and 94.

The bones of Dinosaur 662 were crated on the spot, dragged 20 miles by horse teams to the nearest railroad, and shipped off to Pittsburgh. There, like hundreds of

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Pugh and his wife Karen check molds



The skeleton is 150 million years old

DINOSAUR

continued

tons of other assorted dinosaur fossils collected by Carnegie field parties, they were crammed into the museum basement.

FOR MORE THAN 50 years, Dinosaur 662 lay unassembled in the Carnegie basement. Then, about 1957, it was traded to the Cleveland Museum for a collection of rare mounted birds.

There, it languished unmounted for more years in a museum basement. Then Dr Pulley began trying to get the dinosaur for the expanding Houston museum. It was purchased for \$15,000 raised by the Junior League. Another \$10,000 was set aside for the restoration and assembly by Pugh. The figures do not include the framework and final mounting.

Thus, last January, the remains of Dinosaur 662 arrived in Houston on haul-it-yourself trailers, and Pugh and his two female assistants went to work.

"We called it our giant jigsaw puzzle," Mrs Pugh recalls. "Hundreds of pieces had to be fitted together. I started with little pieces and worked up, but after awhile the pieces got so big I couldn't lift them by myself."

For example, one rear leg bone, the left femur, weighs about 250 pounds.

Some bones were complete and intact. Some were in pieces. Some were partly missing. Some were completely gone.

A rough tabulation showed 36 complete bones and parts or pieces of another 140. Something over 120 bones were missing, notably the ribs, pelvic bones and a section of the caudal vertebrae in the tail.

Even so, Pugh was elated. He believes the Houston dinosaur may be the most nearly complete one in existence.

One strange thing about Dinosaur 662. The vertebrae go along nicely and almost complete up to the third dorsal vertebra, just below the base of the neck. There, from vertebrae almost three feet high, the next six vertebrae drop off to stumps a few inches high. They appear to have been crushed.

"I have a theory that something crushed this animal's back, just behind the neck," Pugh says. "Perhaps it was attacked and killed by a giant carnivorous dinosaur that still is unknown to us."

The crushed vertebrae, of course, were to be rebuilt before the skeleton goes on display in the museum.

THROUGH THE sweltering summer, Mrs Warfield and Mrs Pugh filled in gaps, sculpturing missing bones and grafting missing sections onto the incomplete bones. They used a mixture of sculpting clays and powdered cement for the restoration work.

All pieces then got a gray undercoat, tinges and traces of brown paint, and finally a coat of orange shellac to preserve and harden the bones.

TEMPO

Sometimes Pugh would find one of the brittle old bones cracked in two, even though each bone rested on a styrofoam pillow.

"You never knew whether it was from natural stress or human erosion," Pugh says. "I think we did have quite a human erosion factor."

Through the months, touring groups of Boy Scouts, Girl Scouts, crippled children, Junior Leaguers and school children visited Pugh's workshop at 5809 Southwest Freeway to see the dinosaur taking shape.

Pugh decided that, when the time came, he would transfer the bones to the museum by car, a few pieces at a time, swathed in blankets and foam.

The restorers over the months developed a real affection for their shattered saurian. They nicknamed him—or her—"Dippy."

PUGH, MEANWHILE, was getting more and more hooked on dinosaurs. He learned that Carnegie in 1910 had engaged an Italian named Scarlatti to make a set of bone casts from the 84-94 dinosaur before it was mounted in Pittsburgh. This was done to provide replicas of the diplodocus to eight major museums around the world.

Figuring the casts could be used to fill the gaps in the Houston dinosaur, Pugh traced them after a four-month search to a children's museum in Rocky Mount, NC. At first he was told they had been destroyed.

But through a mixture of diplomacy and detective work, Pugh found them and brought them to his workshop in Houston. He is using them to mold some of the missing parts of the Houston dinosaur.

In another sleuthing job, he set out looking for old plans for diplodocus mounting frameworks. The Pittsburgh museum, he learned, had destroyed their plans and mounting molds.

The search then led him to the British Museum in London, where one of the 1910 replicas had been mounted. There, in the museum basement, the staff found the plans for their mounting and sent them along to Pugh.

Drawn on a rolled-up length of royal linen and labeled in antique script, the British Museum plans will guide Pugh in building a mounting for the Houston dinosaur.

While most motorists whizzing by on the freeway were oblivious to the dinosaur construction inside the barn-like metal building, the project spread ripples of interest through museum and scientific circles around the world. Dr Fuji Takai of the Geological Institute at Tokyo University has corresponded with Pugh. So has G. M. Roding, director of the Natuurhist-Museum in Enschede, Holland.

IT APPEARS NOW that producing dinosaurs may turn into a part-time, lifetime job for Pugh. He says he has had inquiries

from about a dozen museums or institutions interested in getting a dinosaur replica from the Rocky Mount molds.

So when the Houston museum mounting is finished, Pugh plans to recondition the broken and leaky old molds and start turning out full-scale dinosaur kits.

Under an agreement with the Houston museum, Pugh will produce the replicas for the cost of his time and materials, plus a 15 per cent surcharge which will go to the mu-

seum. In some cases the other museums may swap Houston a different type of dinosaur, although Dr Pulley is devoutly hoping "we don't get any more as large as this one."

Pugh is delighted with the plan.

"I'm just vain enough," he says, "to think I can help children all over the world to see these animals and know what they were like. So that's what Arthur Pugh & Family is going to do." ✕



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